A First Course in Abstract Algebra

Section 4: Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

4 Groups . . . 3

Exercise 4.1: Integer multiplication . . . 3

Exercise 4.2: Even integers under addition . . . 3

Exercise 4.3: The geometric-mean operation . . . 3

Exercise 4.4: Rational multiplication including zero . . . 3

Exercise 4.5: Division as an operation . . . 3

Exercise 4.6: The absolute value of a product . . . 3

Exercise 4.7: An abelian group of order one thousand . . . 3

Exercise 4.8: Units modulo eight . . . 4

Exercise 4.9: Distinguishing the circle group . . . 4

Exercise 4.10: Multiples of a positive integer . . . 4

Exercise 4.11: Diagonal matrices under addition . . . 4

Exercise 4.12: All diagonal matrices under multiplication . . . 4

Exercise 4.13: Invertible diagonal matrices . . . 4

Exercise 4.14: Diagonal sign matrices . . . 5

Exercise 4.15: All upper-triangular matrices multiplicatively . . . 5

Exercise 4.16: Upper-triangular matrices additively 5
Exercise 4.17: Upper-triangular matrices of determinant one 5
Exercise 4.18: Matrices of determinant plus or minus one 5
Exercise 4.19: A shifted multiplicative group 5
Exercise 4.20: The groups of order four 6
Exercise 4.21: Counting labeled small group operations 6
Exercise 4.22: Ordering the group axioms 6
Exercise 4.23: Critiquing four proposed definitions 6
Exercise 4.24: Identity and inverses without associativity 7
Exercise 4.25: Ten statements about groups 7
Exercise 4.26: Synopsis of left cancellation 7
Exercise 4.27: Synopsis of solving a group equation 7
Exercise 4.28: Isomorphisms preserve inverses 8
Exercise 4.29: An involution in a group of even order 8
Exercise 4.30: Mixed one-sided axioms 8
Exercise 4.31: Idempotents in a group 8
Exercise 4.32: A group of exponent two is abelian 8
Exercise 4.33: Powers of commuting products 8
Exercise 4.34: A finite-order power 9
Exercise 4.35: A squared-product identity forces commutation 9
Exercise 4.36: When inverses keep their order 9
Exercise 4.37: Cyclically rotating a product equal to the identity 9
Exercise 4.38: Matching left axioms imply the group axioms 9
Exercise 4.39: Solvability of all linear equations 9
Exercise 4.40: The reversed operation 10
Exercise 4.41: Conjugation is an automorphism 10

4 Groups

Exercise 4.1: Integer multiplication

Integer multiplication is associative ($\mathcal{G}_1$) and has identity 1 ($\mathcal{G}_2$). But not every integer has a multiplicative inverse in $\mathbb{Z}$: $0x = 1$ is impossible, and $2x = 1$ has no integer solution. Thus the first failing axiom is $\mathcal{G}_3$, so this is not a group.

Exercise 4.2: Even integers under addition

The sum $2m + 2n = 2(m + n)$ is even, so closure holds. Addition is associative, $0 = 2 \cdot 0$ is the identity, and the inverse of $2m$ is $-2m$, again even. All three group axioms hold, so this is an abelian group.

Exercise 4.3: The geometric-mean operation

Associativity already fails, so $\mathcal{G}_1$ is the first failing axiom. For example, on $\mathbb{R}^+$ let $a = 1, b = 1, c = 16$. Then $(a * b) * c = 1 * 16 = 4$, whereas $a * (b * c) = 1 * 4 = 2$. The positive square roots make this a binary operation, but not a group.

Exercise 4.4: Rational multiplication including zero

Multiplication is associative and has identity 1. Zero has no inverse, since $0q = 0$ for every rational q. Thus $\mathcal{G}_3$ is the first failure and the set $\mathbb{Q}$, as opposed to $\mathbb{Q} \setminus \{0\}$, is not a multiplicative group.

Exercise 4.5: Division as an operation

The operation is closed on nonzero reals, but is not associative. With $a = 1, b = 1, c = 2$, $(a * b) * c = (1/1)/2 = 1/2$, whereas $a * (b * c) = 1/(1/2) = 2$. Therefore the first failing axiom is $\mathcal{G}_1$.

Exercise 4.6: The absolute value of a product

Associativity holds: $(a * b) * c = \big||ab|c\big| = |a||b||c| = \big|a|bc|\big| = a * (b * c)$. However no identity exists in $\mathbb{C}$. Any product $a * e = |ae|$ is a nonnegative real number and so cannot equal, for example, $a = i$. Thus $\mathcal{G}_2$ is the first failure.

Exercise 4.7: An abelian group of order one thousand

Take $\mathbb{Z}_{1000}$ under addition modulo 1000. It has exactly 1000 elements, addition of residues is associative and commutative, zero is the identity, and k has inverse the representative of $-k$ modulo 1000. Hence this example meets every requirement.

Exercise 4.8: Units modulo eight

Multiply each row label by each column label and reduce modulo 8:

$\cdot_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

The table shows closure, identity 1, and that each element is its own inverse. Associativity is inherited from integer multiplication modulo 8. Thus this is an abelian group (the four-element noncyclic group).

Exercise 4.9: Distinguishing the circle group

In U the equation $z^3 = 1$ has three solutions, the three cube roots of unity. In the additive group $\mathbb{R}$, the corresponding equation $3x = 0$ has only $x = 0$. In the multiplicative group $\mathbb{R}^*$, $x^3 = 1$ has only $x = 1$. An isomorphism sends the identity to the identity and preserves products and powers, so would biject these solution sets. Their different sizes rule out both proposed isomorphisms.

Exercise 4.10: Multiples of a positive integer

(a) Sums of multiples of n are multiples of n, addition is associative, $0 = n \cdot 0$ belongs, and the negative of nm is $n(-m)$. Thus $(n\mathbb{Z}, +)$ is a group.

(b) Define $\phi : n\mathbb{Z} \to \mathbb{Z}$ by $\phi(nm) = m$. This is well defined since $n \neq 0$ makes the representation unique; it is bijective with inverse $m \mapsto nm$. Finally $\phi(nm + nk) = m + k = \phi(nm) + \phi(nk)$, so it is an isomorphism.

Exercise 4.11: Diagonal matrices under addition

This is a group. Adding diagonal matrices adds their diagonal entries and leaves off-diagonal entries zero. Associativity follows from matrix addition; the zero matrix is the identity; and negating every diagonal entry gives an additive inverse still in the set. The group is abelian.

Exercise 4.12: All diagonal matrices under multiplication

The set is closed, multiplication is associative, and I_n is its identity, but the zero matrix has no multiplicative inverse. Therefore it is not a group. Requiring nonzero diagonal entries would remove this obstruction.

Exercise 4.13: Invertible diagonal matrices

The product of $\operatorname{diag}(a_1, \ldots, a_n)$ and $\operatorname{diag}(b_1, \ldots, b_n)$ is $\operatorname{diag}(a_1b_1, \ldots, a_nb_n)$ with nonzero diagonal. The identity is I_n, the inverse is $\operatorname{diag}(1/a_1, \ldots, 1/a_n)$, and multiplication is associative. Thus these matrices form an abelian group.

Exercise 4.14: Diagonal sign matrices

A product of two signs is a sign, so these diagonal matrices are closed. The identity has all diagonal entries 1, and each matrix is its own inverse since $(\pm 1)^2 = 1$. Associativity is inherited from matrix multiplication. Hence this is a group of 2^n elements.

Exercise 4.15: All upper-triangular matrices multiplicatively

Products remain upper triangular and I_n belongs, but singular upper-triangular matrices are also included. In particular the zero matrix cannot have a multiplicative inverse. Thus the given set is not a group under multiplication.

Exercise 4.16: Upper-triangular matrices additively

Sums and negatives remain upper triangular. Matrix addition is associative, the zero matrix is an additive identity in this set, and $-A$ is the additive inverse of A. Therefore these matrices form an abelian group under addition.

Exercise 4.17: Upper-triangular matrices of determinant one

If A, B are upper triangular with determinant 1, so is AB, since $\det(AB) = 1$. The identity belongs. Each A is invertible, and A^{-1} is upper triangular: solving $Ax = e_j$ by back substitution gives zeros below the jth position in column j of the inverse. Also $\det(A^{-1}) = 1$. Associativity is inherited, so this is a group.

Exercise 4.18: Matrices of determinant plus or minus one

The determinant of a product is the product of determinants, so closure holds in $\{1, -1\}$. Every such matrix is invertible, I_n belongs, and $\det(A^{-1}) = 1/\det A \in \{1, -1\}$. Associativity is matrix associativity, so the indicated set is a group.

Exercise 4.19: A shifted multiplicative group

Use the identity $a * b + 1 = (a + 1)(b + 1)$.

(a) If $a, b \neq -1$, their shifted values are nonzero, so $(a * b) + 1 \neq 0$ and $a * b \neq -1$. The formula specifies one real output, proving it is a binary operation on S.

(b) The bijection $\phi : S \to \mathbb{R}^*$, $\phi(a) = a + 1$, satisfies $\phi(a * b) = \phi(a)\phi(b)$, so it transports the group laws from $\mathbb{R}^*$. Explicitly, both bracketings have shifted value $(a + 1)(b + 1)(c + 1)$, proving associativity; the identity is 0, and the inverse of a is $-a/(a+1)$, whose shifted value is $1/(a+1)$.

(c) Apply ϕ to $2 * x * 3 = 7$: $3(x + 1)4 = 8$. Thus $x + 1 = 2/3$ and $x = -1/3 \in S$. Checking directly, $2 * (-1/3) = 1$ and $1 * 3 = 7$.

Exercise 4.20: The groups of order four

No repeated entry can occur in a row or column of a group table, by cancellation. Filling accordingly gives the following three tables in the order e, a, b, c:

T_1	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

T_2	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

T_3	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

The first two have $a^2 = e$; the third has $a^2 = b$. In T_2, b has successive powers e, b, a, c, whereas in T_3, a has successive powers e, a, b, c. Therefore $T_2 \cong T_3$ via $e \mapsto e, a \mapsto b, b \mapsto a, c \mapsto c$. In T_1 every nonidentity element squares to e, so it cannot be isomorphic to either cyclic table.

(a) Yes: all three tables are symmetric, so every group of order four is abelian.

(b) T_3 is U_4 via $e \mapsto 1, a \mapsto i, b \mapsto -1, c \mapsto -i$; T_2 is also U_4 via its preceding relabeling.

(c) T_1 is the diagonal-sign group for $n = 2$: send $a \mapsto \mathrm{diag}(-1, 1)$, $b \mapsto \mathrm{diag}(1, -1)$, $c \mapsto -I_2$. Multiplying these matrices verifies the table and therefore its associativity.

Exercise 4.21: Counting labeled small group operations

On a fixed two-element set there are two group operations: either element can be the identity, and that choice forces the other element to square to it. On a fixed three-element set there are three operations, one for each choice of identity; after that choice the two other elements must be inverses and their squares are forced. Swapping those two nonidentity labels leaves the same cyclic group table, so does not create an additional operation. Thus the answers are 2 and 3.

Exercise 4.22: Ordering the group axioms

The inverse axiom $\mathcal{G}_3$ refers to the identity e introduced in $\mathcal{G}_2$. Thus $\mathcal{G}_2$ must precede $\mathcal{G}_3$. The unacceptable orders are $\mathcal{G}_1\mathcal{G}_3\mathcal{G}_2$, $\mathcal{G}_3\mathcal{G}_1\mathcal{G}_2$, and $\mathcal{G}_3\mathcal{G}_2\mathcal{G}_1$. The three orders with $\mathcal{G}_2$ first relative to $\mathcal{G}_3$ are acceptable; associativity has no such dependence on the other two.

Exercise 4.23: Critiquing four proposed definitions

A correct definition specifies a binary operation on G, associativity, an element $e \in G$ satisfying $ex = xe = x$ for every x, and for each $a \in G$ an $a^{-1} \in G$ satisfying $aa^{-1} = a^{-1}a = e$.

(a) The equality ending "x = identity" is incorrect: x is arbitrary, not necessarily the identity. The identity equation needs "for every $x \in G$," the inverse must belong to G, and the same operation symbol must be used consistently.

(b) It refers to "the operation" without specifying a binary operation on G. It should define what the identity and inverse equations mean and explicitly put each inverse in G.

(c) Associativity is missing. "An inverse exists" must mean an inverse for every element, relative to the specified identity; merely defining a binary operation is already part of the setup.

(d) "Associative under addition" confuses the operation with another operation. The notation $\{e\}$ denotes a set, not the element e; the identity equation must end in a, not e. The inverse must be in G, and the universal quantifiers must govern the appropriate equations.

Exercise 4.24: Identity and inverses without associativity

Consider

$*$	e	a	b
e	e	a	b
a	a	e	e
b	b	e	e

The first row and column show e is a two-sided identity. Both a and b are their own two-sided inverses, so $\mathcal{G}_2, \mathcal{G}_3$ hold. But $(a * a) * b = e * b = b$, whereas $a * (a * b) = a * e = a$. Thus associativity fails, as requested.

Exercise 4.25: Ten statements about groups

(a) False: two identities e, f satisfy $e = ef = f$.

(b) True: the cancellation rule forces the unique three-element cyclic table, up to relabeling.

(c) True for the linear equations in this section: $ax = b$ and $ya = b$ have the solutions $a^{-1}b$ and ba^{-1}.

(d) False: understanding and stating the exact content matters; identical wording is unnecessary.

(e) False: one implication could allow a definition that excludes some actual groups. Equivalence requires both directions.

(f) True: equivalence of the two definitions is precisely what is needed.

(g) True: the one-, two-, and three-element group tables are all commutative.

(h) True: $axb = c$ has the unique solution $x = a^{-1}cb^{-1}$, obtained by multiplying on the indicated sides.

(i) False: a group must contain its identity, so it is nonempty.

(j) True: a group is a set with a binary operation satisfying additional axioms.

Exercise 4.26: Synopsis of left cancellation

Multiply an equality with the same left factor on the left by that factor's inverse, then reassociate and use the identity to remove that factor.

Exercise 4.27: Synopsis of solving a group equation

Multiplying the desired right-hand side on the left by the inverse of the coefficient produces a solution. Left cancellation shows that any two solutions must agree.

Exercise 4.28: Isomorphisms preserve inverses

Let a^{-1} be the inverse of a and let e' be the target identity. Since $\phi(e) = e'$,

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = e', \qquad \phi(a^{-1})\phi(a) = \phi(a^{-1}a) = e'.$$

Thus $\phi(a^{-1})$ is a two-sided inverse of $\phi(a)$. The inverse of a group element is unique, so $\phi(a^{-1}) = \phi(a)^{-1}$.

Exercise 4.29: An involution in a group of even order

Pair each element x with its inverse. Elements with $x \neq x^{-1}$ occur in disjoint pairs and hence contribute an even number of elements. If e were the only self-inverse element, the total size would therefore be even plus one, which is odd. Since the group has even size, there must be some $a \neq e$ with $a = a^{-1}$, equivalently $a^2 = e$.

Exercise 4.30: Mixed one-sided axioms

On $\mathbb{R}^*$ put $a * b = |a|b$.

(a) Outputs are nonzero, and $(a * b) * c = ||a|b|c = |a||b|c = a * (b * c)$, proving associativity.

(b) The element 1 is a left identity, since $1 * b = b$. Each a has a right inverse relative to 1, namely $b = 1/|a|$, since $a * b = 1$.

(c) This is not a group. A right identity r would need $1 * r = 1$, hence $r = 1$, but then $(-1) * r = 1 \neq -1$.

(d) A left identity combined with right inverses does not suffice for the group axioms. The valid one-sided alternatives require the identity and inverse axioms to have matching sides (both left, or both right). This example even has two left identities, 1 and -1.

Exercise 4.31: Idempotents in a group

The identity is idempotent since $e^2 = e$. If $x^2 = x$, multiply on the left by x^{-1} to obtain $x = e$. Thus every idempotent equals the identity, and there is exactly one.

Exercise 4.32: A group of exponent two is abelian

Assume $x^2 = e$ for every $x \in G$. In particular $(ab)^2 = e$, so $abab = e$. Multiplying this equality on the left by a and on the right by b gives $ba = ab$, because $a^2 = b^2 = e$. Since a, b were arbitrary, the group is abelian. No commutation was used before establishing the conclusion.

Exercise 4.33: Powers of commuting products

Fix a, b in an abelian group. For $n = 1$, $(ab)^1 = a^1b^1$. If $(ab)^n = a^nb^n$, then

$$(ab)^{n+1} = (ab)^n ab = a^nb^nab = a^nab^nb = a^{n+1}b^{n+1}.$$

Commutativity permits moving a past b^n, and associativity permits regrouping. Thus mathematical induction proves the identity for every positive integer n.

Exercise 4.34: A finite-order power

Let $|G| = m$. The $m + 1$ powers $e, a, a^2, \ldots, a^m$ cannot all be distinct, so $a^i = a^j$ for some $0 \leq i < j \leq m$. Multiply by a^{-i} to get $a^{j-i} = e$. Since $j - i > 0$, this supplies the required positive exponent.

Exercise 4.35: A squared-product identity forces commutation

The hypothesis is $abab = aabb$. Multiply on the left by a^{-1} to get $bab = abb$, then on the right by b^{-1} to get $ba = ab$. The sides of multiplication matter; cancellation is valid without assuming the group is abelian.

Exercise 4.36: When inverses keep their order

In any group $(ab)^{-1} = b^{-1}a^{-1}$, since this element gives e on multiplication on either side. If $ab = ba$, then their inverses also commute, so this equals $a^{-1}b^{-1}$. Conversely, if $(ab)^{-1} = a^{-1}b^{-1}$, comparison with the general formula gives $b^{-1}a^{-1} = a^{-1}b^{-1}$. Taking inverses reverses both products and yields $ab = ba$. This proves both implications.

Exercise 4.37: Cyclically rotating a product equal to the identity

From $abc = e$, multiply on the left by a^{-1} to get $bc = a^{-1}$. Multiplying on the right by a then gives $bca = a^{-1}a = e$. This is a cyclic rotation, not an arbitrary interchange of factors.

Exercise 4.38: Matching left axioms imply the group axioms

Assume associativity, a left identity e, and a left inverse a' for every a.

First obtain left cancellation.

If $ax = ay$, multiply on the left by a'. Then $(a'a)x = (a'a)y$ gives $ex = ey$, hence $x = y$.

Prove the identity is also right-sided.

Since $a'(ae) = (a'a)e = ee = e = a'a$, left cancellation of a' gives $ae = a$.

Prove each left inverse is also right-sided.

Now $a'(aa') = (a'a)a' = ea' = a' = a'e$. Cancel a' on the left to get $aa' = e$. Thus e is a two-sided identity and each a' is a two-sided inverse. All ordinary group axioms hold.

Exercise 4.39: Solvability of all linear equations

Choose $a \in G$, possible since G is nonempty. By the solvability hypothesis choose e with $ea = a$. For any $g \in G$, choose z with $az = g$. Associativity gives $eg = e(az) = (ea)z = az = g$, so e is a left

identity for every element. For any $c \in G$, solvability of $yc = e$ gives a left inverse c' with $c'c = e$. Thus the matching left axioms of Exercise 4.38 hold, and its proof shows that G is a group.

Exercise 4.40: The reversed operation

Write the original multiplication as juxtaposition and set $a * b = ba$. Then $(a * b) * c = cba = a * (b * c)$, the original identity remains an identity, and the original a^{-1} is also the $*$-inverse. Thus $(G, *)$ is a group. Define $\phi(a) = a^{-1}$. It is bijective because applying it twice gives a, and

$$\phi(a * b) = \phi(ba) = (ba)^{-1} = a^{-1}b^{-1} = \phi(a)\phi(b).$$

Hence it is an isomorphism from the reversed group to the original group.

Exercise 4.41: Conjugation is an automorphism

For fixed $g \in G$ define $i_g(x) = gxg^{-1}$. Its inverse map is $i_{g^{-1}}$, since $g^{-1}(gxg^{-1})g = x$ and the reverse composition also equals the identity. Thus i_g is bijective. Moreover

$$i_g(x)i_g(y) = (gxg^{-1})(gyg^{-1}) = gx(g^{-1}g)yg^{-1} = gxyg^{-1} = i_g(xy).$$

Therefore conjugation preserves multiplication and is an isomorphism of G with itself.